M378K: 03/04/2026.

The $\chi^2(n)$-distribution.

- If $Y \sim N(0,1)$, then $\boxed{W = Y^2 \sim \chi^2(df=1)}$

  Its density is
  $$f_W(w) = \frac{1}{\sqrt{2\pi w}} \cdot e^{-\frac{w}{2}} \cdot \mathbb{1}_{(0,\infty)}(w)$$

  Its mgf is
  $$\boxed{m_W(t) = (1-2t)^{-1/2} = \frac{1}{\sqrt{1-2t}}}$$

- $Y_1 \sim N(0,1)$, $Y_2 \sim N(0,1)$ independent

  Set $\boxed{W = Y_1^2 + Y_2^2} \sim \chi^2(df=2)$

  Let's find its mgf:
  $$m_W(t) = m_{Y_1}(t) \cdot m_{Y_2}(t)$$
  $$= \frac{1}{\sqrt{1-2t}} \cdot \frac{1}{\sqrt{1-2t}} = \frac{1}{1-2t}$$

  From the HW, we know that $\boxed{W \sim E(\tau=2)}$

  $\boxed{\text{So, } \chi^2(df=2) \text{ is the same as } E(\tau=2).}$

Def'n. The χ^2-dist w/ n degrees of freedom is the dist'n of the sum
$$W = Y_1^2 + Y_2^2 + \cdots + Y_n^2$$
where
$$Y_i \sim N(0,1) \text{ for } i=1..n \text{ an independent.}$$
We write $\boxed{W \sim \chi^2(df=n) = \chi^2(n)}$

Note: $m_W(t) = \left(\frac{1}{\sqrt{1-2t}}\right)^n = (1-2t)^{-\frac{n}{2}} = \left(\frac{1}{1-2t}\right)^{\frac{n}{2}}$

The Gamma Distribution.

Def'n. A random variable Y is said to have the **gamma distribution** w/ parameters $k>0$ and $\tau>0$ if its mgf is of the form

$$m_Y(t) = \left(\frac{1}{1-\tau t}\right)^{k}$$

We write $\boxed{Y \sim \Gamma(k, \tau)}$.

scale parameter

k... shape parameter

Note:

$$\boxed{\begin{aligned} \mathbb{E}[Y] &= k \cdot \tau \\ Var[Y] &= k \cdot \tau^2 \end{aligned}}$$

Q: Say that $Y \sim \Gamma(1, \tau)$. What's another name for it?

$Y \sim E(\tau)$ ☺

Q: Say that $Y \sim \Gamma(\frac{n}{2}, 2)$. What's another name for it?

$Y \sim \chi^2(n)$ ☺

Q: $Y_1 \sim \Gamma(k_1, \tau)$, $Y_2 \sim \Gamma(k_2, \tau)$ independent

$$\boxed{Y_1 + Y_2 \sim \Gamma(k_1 + k_2, \tau)}$$

The Statistical Set-Up.

Population. e.g., all the people in this class

Sample. e.g., a committee of 4 students

We assume that it's a **REPRESENTATIVE SAMPLE.**

Use the same word for the results of measuring (or polling) from the population w/ an unknown but common dist'n.

Def'n. A random sample of size n from a distribution D is a random vector

$$(Y_1, Y_2, \dots, Y_n)$$

such that:

1. $Y_1, \dots, Y_n$ are independent
2. every Y_i has the distribution D.

Example. Consider 10 measurement $Y_1, Y_2, \dots, Y_{10}$. Care was taken so that they're independent.

It's a standard model to assume that

Y_i are normally distributed w/ an unknown mean μ.

Scenario #1. We know the standard deviation 0.1.

Then, $Y_i \sim N(\mu, \sigma = 0.1)$, $i = 1..10$

Scenario #2. We don't know the standard deviation σ.

Then, $Y_i \sim N(\mu, \sigma)$, $i = 1..10$

Def'n. Any function of the random sample is called a **STATISTIC**.

A **POINT ESTIMATOR** is any function (rule, procedure) of the random sample $(Y_1, \dots, Y_n)$ which included only **known** constants (w/ the purpose of estimating a model parameter).

An **interval estimator** is a pair of point estimators.

ESTIMATORS MUSTN'T CONTAIN THE UNKNOWN PARAMETER WE'RE TRYING TO ESTIMATE.

e.g.,

- $\dfrac{\text{\# of people in the "committee" who like ice cream}}{4}$,

  i.e., a sample proportion;

- in the normal example, we look @ the sample mean

$$\boxed{\bar{Y} = \frac{Y_1 + Y_2 + \dots + Y_n}{n}}$$